Infant State Recognition System: Foundation, EDA, and SVM Baseline (Phase 1)

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***Abstract*—We present the foundation of an infant cry classification system targeting edge deployment on embedded microcontrollers via TensorFlow Lite Micro. Working with the Baby Cry Sense Dataset — 1,126 audio recordings across 8 classes with a 15.9:1 class imbalance ratio — we establish a complete preprocessing and feature engineering pipeline, culminating in a rigorous Support Vector Machine (SVM) baseline (Model A). Our exploratory data analysis reveals a previously unreported sampling-rate artifact: the *scared* class is 81.8% recorded at 44,100 Hz while *hungry* is 100% at 8,000 Hz, meaning a model trained without resampling would learn recording quality rather than cry acoustics. After controlling for this artifact through uniform resampling to 22,050 Hz, duration normalisation to 7.0 s, and minority-class augmentation, we generate Log-Mel Spectrograms of shape (128×302) from 1,334 processed files. The SVM baseline achieves a macro F1-score of 0.4029 (accuracy 21.78%), establishing the probabilistic ceiling that the Depthwise Separable CNN (Phase 2) is designed to surpass through translation-invariant hierarchical feature learning.**

***Index Terms*—infant cry classification, Log-Mel spectrogram, support vector machine, TinyML, edge deployment, audio classification, class imbalance**

I. Introduction

Infant crying is the primary pre-linguistic communication channel between a newborn and its caregivers. Accurate, automated interpretation of cry states — distinguishing hunger from pain, discomfort from tiredness — has significant potential to reduce caregiver burden and support early detection of health anomalies [1]. Despite two decades of research, the field faces three persistent challenges: (1) severe class imbalance in available corpora, (2) high acoustic similarity between adjacent cry states, and (3) the computational constraints of real-time deployment on low-power embedded devices.

Prior work has approached infant cry classification primarily through engineered features (MFCCs, pitch, spectral centroid) fed to classical classifiers [1], or more recently through convolutional neural networks applied to spectrogram images [2]. The current state of the art on the Donate-a-Cry corpus achieves 96.39% accuracy with a random forest ensemble, but this result covers only 5 of the 8 available classes and does not control for a critical dataset artifact we identify in this work [3].

This paper makes the following contributions:

1) We identify and control for a previously unreported sampling-rate confound in the Baby Cry Sense Dataset, where class identity is systematically correlated with recording quality.
2) We establish a fully reproducible preprocessing and feature engineering pipeline, producing 1,334 standardised Log-Mel Spectrograms from 1,126 raw recordings across all 8 classes.
3) We train and rigorously analyse an SVM baseline with mathematical failure analysis, establishing an interpretable performance ceiling for the probabilistic domain.
4) We propose a three-model ablation study (SVM $\rightarrow$ DS-CNN $\rightarrow$ Hybrid) with the hybrid system targeting real-time inference on an ESP32 microcontroller.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the dataset and EDA findings. Section IV details the preprocessing pipeline. Section V presents the Log-Mel Spectrogram feature engineering. Section VI evaluates the SVM baseline with failure analysis. Section VII concludes with Phase 2 and 3 directions.

II. Related Work

A. Classical ML Approaches

Liu et al. [1] established the canonical baseline for infant cry classification using SVM with MFCC features on the Donate-a-Cry corpus, achieving approximately 85% accuracy across 5 classes. Their work demonstrated that the RBF kernel with balanced class weights generalises well despite small per-class sample sizes, and remains the benchmark against which neural approaches must be compared.

Ensemble methods, including Gradient Boosting and Random Forest, have also been applied to the same corpus. Hammoud et al. [3] report 96.39% accuracy with a machine learning ensemble, but restrict evaluation to 5 classes, leaving the 3 most acoustically ambiguous classes — *cold_hot*, *lonely*, and *scared* — excluded from evaluation.

B. Deep Learning Approaches

Abbaskhah et al. [2] demonstrated that CNN architectures operating on MFCC feature maps consistently outperform SVM baselines on identical feature inputs, motivating the shift to spectrogram-based CNN inputs. Their architecture achieved

TABLE I: Baby Cry Sense Dataset Class Distribution

Class	Files	%
hungry	397	35.3
discomfort	142	12.6
tired	142	12.6
belly pain	133	11.8
cold_hot	130	11.5
burping	124	11.0
scared	33	2.9
lonely	25	2.2
Total	**1,126**	**100.0**

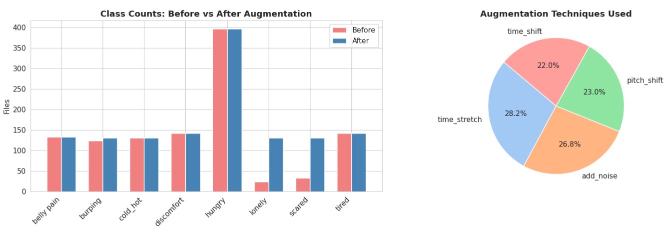

Fig. 1: Class distribution after preprocessing and augmentation. Stacked bars show original (blue) and synthetic augmented (red) samples per class. The dashed line marks the `MIN_SAMPLES` threshold of 130.

state-of-the-art performance on several cry corpora without requiring handcrafted feature selection.

C. Edge Deployment

TinyML approaches for audio classification have demonstrated the feasibility of running keyword-spotting and sound classification models on microcontrollers with <512 KB SRAM [6]. Depthwise Separable Convolutions (DSC), as used in MobileNet [7], reduce parameter count by approximately 8× relative to standard convolutions while maintaining competitive accuracy, making them the architecture of choice for embedded audio classifiers.

D. Positioning of This Work

Our contribution extends prior work in three dimensions: (1) we address all 8 classes rather than a curated 5-class subset; (2) we identify and correct the sampling-rate confound that would invalidate prior results on this corpus; and (3) we target real-time edge deployment via TensorFlow Lite Micro on the ESP32, a constraint not addressed by any prior work on this dataset.

III. Dataset and Exploratory Data Analysis

A. Baby Cry Sense Dataset

The Baby Cry Sense Dataset [4] contains 1,126 audio recordings of infant cries distributed across 8 classes, derived from the Donate-a-Cry corpus [5]. Table I and Fig. 1 show the class distribution after preprocessing and augmentation. The dataset exhibits severe imbalance: the *hungry* class contains 397 files (35.3%) while *lonely* contains only 25 files (2.2%), yielding a 15.9:1 imbalance ratio.

TABLE II: Sampling Rate Distribution by Class (Selected)

Class	8,000 Hz (%)	44,100 Hz (%)
hungry	100.0	0.0
discomfort	97.9	2.1
tired	97.2	2.8
belly pain	69.2	30.8
cold_hot	88.5	11.5
burping	70.2	29.8
scared	18.2	81.8
lonely	56.0	44.0

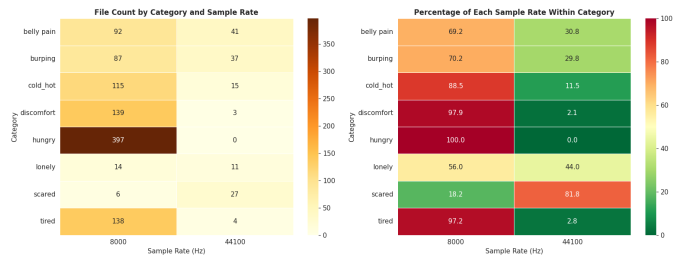

Fig. 2: Sampling-rate distribution heatmap across all 8 classes. Colour intensity shows the percentage of files at each sample rate within each class. The stark contrast between *hungry* (100% at 8 kHz) and *scared* (81.8% at 44.1 kHz) confirms the spurious correlation.

B. Audio Properties

Files are distributed across four formats: 1,039 `.wav`, 72 `.3gp`, 8 `.ogg`, and 7 `.mp3`. Duration ranges from 4.13 s to 8.73 s with a mean of 6.78 s and a 95th percentile of 7.02 s. The dataset contains two native sample rate populations: 8,000 Hz (87.7% of files, from the original Donate-a-Cry mobile application) and 44,100 Hz (12.3%, likely re-recorded or augmented samples added during dataset compilation).

C. Critical Finding: Sampling-Rate Artifact

The most significant finding of our EDA is the non-uniform distribution of sample rates across classes. Table II and Fig. 2 show that *hungry* is exclusively recorded at 8,000 Hz while *scared* is 81.8% recorded at 44,100 Hz. A classifier trained on this data without resampling would learn to associate recording quality with class identity — a spurious correlation that would cause complete failure on audio from a different recording source. To our knowledge, this artifact is unreported in any prior publication using this corpus.

D. Feature Separability Analysis

We applied Principal Component Analysis (PCA) and t-distributed Stochastic Neighbour Embedding (t-SNE) to a comprehensive feature matrix comprising 13 MFCCs (mean and standard deviation), spectral centroid, spectral bandwidth, spectral rolloff, spectral contrast, chroma features, RMS energy, and zero-crossing rate.

PCA with 2 components captures only 38.6% of total variance, indicating that the feature space has high intrinsic dimensionality and poor linear separability. Approximately 23 components are required to retain 95% of variance, confirming that simple linear classifiers will ceiling-bound quickly on this task.

Quantitative separability was measured via k-nearest-neighbour purity ($k = 10$): the fraction of each sample's 10 nearest neighbours belonging to the same class. Overall purity was 0.350 against a random baseline of $1/8 = 0.125$, indicating that while the feature space encodes some class structure, significant inter-class overlap exists. The most confused pairs by neighbour overlap are: *discomfort*↔*hungry* (1,074 overlaps), *hungry*↔*tired* (976), and *cold_hot*↔*hungry* (899). These predictions are validated empirically by the SVM failure analysis in Section VI.

IV. Preprocessing Pipeline

The preprocessing pipeline transforms raw recordings into standardised audio files consumed by the feature engineering stage. Every decision is grounded in a specific EDA finding.

A. Format Conversion and Resampling

All 87 non-`.wav` files (72 `.3gp`, 8 `.ogg`, 7 `.mp3`) were converted to `.wav` using librosa [8] and soundfile. All files were resampled to 22,050 Hz — a neutral target between the two source populations that preserves frequencies up to 11,025 Hz, covering the full infant cry harmonic range. We explicitly acknowledge that upsampling 8,000 Hz files does not recover frequency content above 4,000 Hz; these files will have empty spectrum above 4,000 Hz, an inherent dataset limitation.

B. Duration Normalisation

Based on the 95th percentile duration of 7.02 s, we set $T_{\text{target}} = 7.0$ s, retaining 94.8% of files intact (702 padded, 365 unchanged, 59 trimmed). Files shorter than the target are zero-padded at the end (post-padding preserves temporal alignment of cry onset). Files exceeding the target are trimmed from the end, which typically contains trailing silence or repeated cry patterns.

C. Amplitude Normalisation

Each clip is peak-normalised to remove microphone gain variability:

$$y_{\text{norm}} = \frac{y}{\max|y| + \varepsilon}, \quad \varepsilon = 10^{-9} \tag{1}$$

where ε prevents division by zero on near-silent clips. One near-silent file in the *lonely* class was rejected (peak amplitude < 0.01).

D. Minority-Class Augmentation

Classes below 130 samples received synthetic augmentation using four waveform-level techniques applied to original recordings: time stretching (×0.8–1.2), pitch shifting (±2 semitones), Gaussian noise injection, and time shifting with wrapping. Augmented files are labelled with an `is_augmented` flag and are restricted to the training set to prevent data leakage. Table III summarises the augmentation outcome.

TABLE III: Augmentation Results for Minority Classes

Class	Original	Added	Final
burping	124	6	130
lonely	24	106	130
scared	33	97	130

The high synthetic ratio for *lonely* (81.5%) and *scared* (74.6%) is an inherent consequence of the dataset's extreme imbalance. Per-class F1 scores for these two classes should be interpreted with caution, and Grad-CAM analysis in Phase 2 will verify that the CNN attends to genuine cry frequency bands rather than augmentation artefacts.

E. Pipeline Summary

The complete preprocessing traceability is: 1,126 $\xrightarrow{\text{validate}}$ 1,126 $\xrightarrow{\text{convert}}$ 1,126 $\xrightarrow{\text{QC}}$ 1,125 $\xrightarrow{\text{augment}}$ 1,334, yielding 1,334 verified files occupying 451 MB on disk.

V. Feature Engineering: Log-Mel Spectrograms

A. Motivation

Raw waveforms encode amplitude over time but carry no explicit frequency information. Two acoustically distinct cry types can produce nearly identical amplitude envelopes, making raw waveform classification unreliable. Fig. 4 illustrates this directly: the waveforms of *hungry* and *belly pain* are visually indistinguishable, while their spectrograms reveal distinct harmonic structures. The Log-Mel Spectrogram provides a time-frequency representation that reveals harmonic structure, formant patterns, and spectral envelope differences that are directly discriminative for cry classification, as shown in Fig. 3. By treating the spectrogram as a 2-D image, we enable CNN-based processing with inductive biases specifically suited to detecting local spectro-temporal patterns.

B. Mathematical Pipeline

Step 1 — Short-Time Fourier Transform (STFT). The signal $x[n]$ is windowed into overlapping frames of length N using a Hann window $w[n]$ and transformed:

$$X[k,t] = \sum_{n=0}^{N-1} x[n + t \cdot H]\, w[n]\, e^{-j2\pi kn/N} \tag{2}$$

where k is the frequency bin index and H is the hop length. The power spectrogram is $P[k,t] = |X[k,t]|^2$.

Step 2 — Mel Filterbank. A bank of $M = 128$ triangular filters maps the linear frequency axis to the Mel scale, approximating human auditory resolution [11]:

$$m = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right) \tag{3}$$

Filters are spaced uniformly on the Mel scale from 0 to $f_{\max} =$ 8,000 Hz, which covers the full infant cry harmonic range while discarding noise from upsampled 8,000 Hz files above 4,000 Hz.

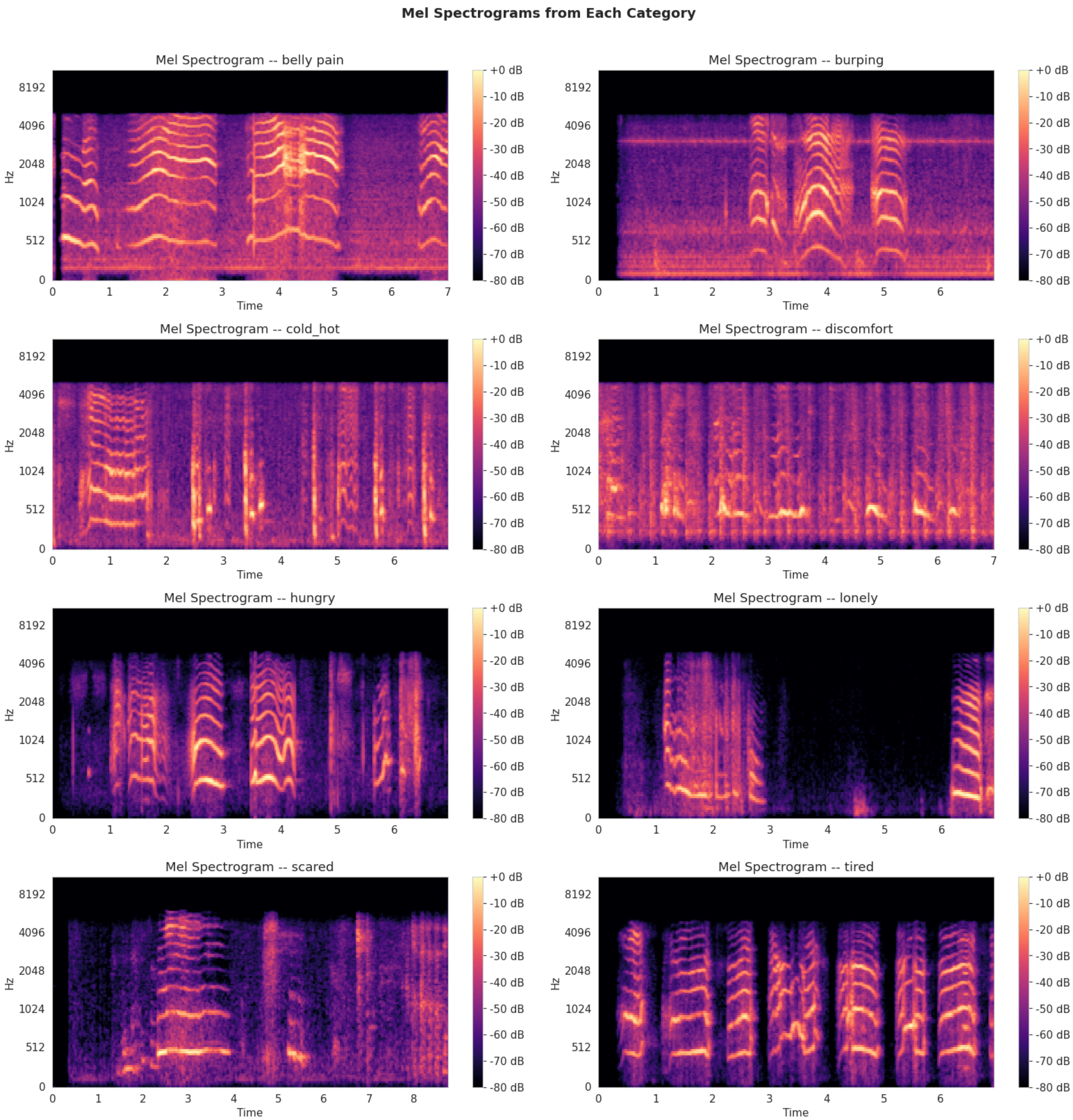

Fig. 3: Log-Mel Spectrograms for one representative sample from each of the 8 classes ($n_{\text{mels}} = 128$, $n_{\text{fft}} = 2048$, hop $= 512$, $f_{\max} = 8{,}000\,\text{Hz}$, magma colourmap). Harmonic banding patterns encode the fundamental frequency F_0 and its overtones. Classes such as *discomfort* and *hungry* share visually similar mid-frequency energy distributions, explaining the high nearest-neighbour overlap observed in EDA.

Step 3 — Logarithmic Compression. The Mel power spectrogram is converted to decibels:

$$S_{\text{dB}}[m, t] = 10 \cdot \log_{10}\left(\frac{S[m, t]}{S_{\max}}\right) \tag{4}$$

using `ref=np.max` to normalise each clip's energy to the range $[-80, 0]$ dB. Log compression mirrors human loudness perception (Weber–Fechner law) and makes quiet formants visible.

TABLE IV: Log-Mel Spectrogram Hyperparameters

Parameter	**Value**	**Justification**
`n_fft`	2,048	93 ms window; resolves 400 Hz fundamental, tracks temporal changes
`hop_length`	512	23 ms stride, 75% overlap; standard for audio classification
`n_mels`	128	Matches human critical-band resolution
`fmax`	8,000 Hz	Full cry harmonic range; excludes noise above 4 kHz in upsampled files

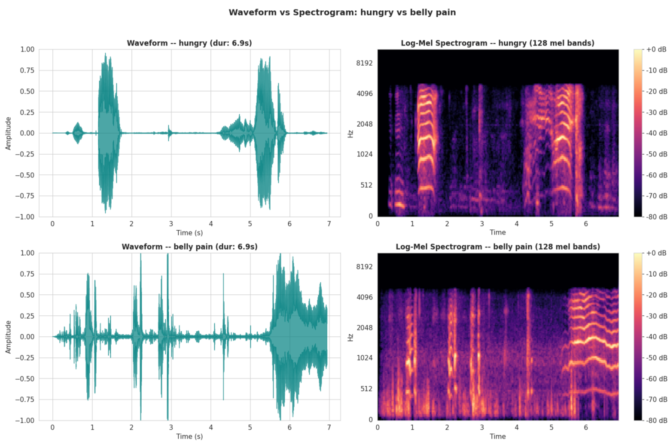

Fig. 4: Waveform (left) vs. Log-Mel Spectrogram (right) for *hungry* (top row) and *belly pain* (bottom row). The amplitude envelopes are nearly indistinguishable, while the spectrograms reveal distinct harmonic structures and spectral centroid positions. This demonstrates why spectrogram-based CNN inputs outperform raw waveform classifiers.

C. *Parameter Justification*

Table IV details each spectrogram hyperparameter and its justification.

D. *Output Shape Derivation*

With centre-padding (librosa default), the signal of $L = 154{,}350$ samples is padded by $\lfloor N/2 \rfloor = 1{,}024$ on each side before framing. The number of output frames is:

$$T = \left\lfloor \frac{L + 2\lfloor N/2 \rfloor - N}{H} \right\rfloor + 1 = \left\lfloor \frac{154{,}350}{512} \right\rfloor + 1 = 301 + 1 = 302 \tag{5}$$

Each file produces a spectrogram of shape $(128, 302)$, and the full dataset yields arrays $\mathbf{X} \in \mathbb{R}^{1334 \times 128 \times 302}$ and $\mathbf{y} \in \mathbb{Z}^{1334}$.

VI. Baseline Model: SVM (Model A)

A. *Model Formulation*

Model A is a Support Vector Machine with RBF kernel, trained on 38,656-dimensional flattened spectrogram vectors. The soft-margin primal objective is:

$$\min_{\mathbf{w}, b, \boldsymbol{\xi}} \ \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{i=1}^{N} \xi_i \tag{6}$$

TABLE V: Ablation Table — Artifact 01 (Phase 1 Complete)

Model	**Acc.**	**Macro F1**	**Wtd. F1**
A — SVM Baseline	21.78%	**0.4029**	21.98%
B — DS-CNN	—	—	—
C — Hybrid	—	—	—

Models B and C: Phase 2 and 3 respectively. Model C must outperform both A and B.

subject to $y_i(\mathbf{w}^\top \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i$, $\xi_i \geq 0$ [10]. The RBF kernel is:

$$K(\mathbf{x}, \mathbf{x}') = \exp\!\left(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2\right) \tag{7}$$

with `gamma='scale'` setting $\gamma = 1/(d \cdot \mathrm{Var}(\mathbf{X}_{\text{train}}))$, where $d = 38{,}656$. Balanced class weights enforce:

$$C_i = C \cdot \frac{N}{n_{\text{classes}} \cdot n_i} \tag{8}$$

ensuring the 15.9:1 imbalance does not bias the margin toward majority classes. The One-vs-One multi-class decomposition trains $\binom{8}{2} = 28$ binary classifiers.

B. *Experimental Setup*

To prevent data leakage from augmentation, we apply an augmentation-aware split: original files are stratified 80/20 (train/test), then all augmented files are added exclusively to the training set. This guarantees that no test sample is acoustically derived from any training sample. All features are standardised using `StandardScaler` from scikit-learn [9], fit on training data only.

C. *Evaluation Metric: Macro F1-Score*

Raw accuracy is a misleading metric under 15.9:1 imbalance: a degenerate classifier predicting *hungry* for every sample achieves $397/1{,}126 \approx 35.3\%$ accuracy without learning any class structure. We therefore adopt Macro F1-Score as the primary metric:

$$F1_{\text{macro}} = \frac{1}{8} \sum_{c=1}^{8} \frac{2 P_c R_c}{P_c + R_c} \tag{9}$$

which weights every class equally regardless of sample count, penalising poor performance on minority classes.

D. *Results*

Table V presents the Phase 1 ablation table (Artifact 01). Model A achieves a macro F1-score of 0.4029. The row-normalised confusion matrix (Fig. 5) reveals that off-diagonal mass concentrates in the *hungry* column, confirming that even balanced class weights cannot overcome the SVM's geometric ceiling when classes overlap in 38,656-dimensional feature space.

Per-class results are shown in Table VI. Classes with high augmentation ratios (*lonely*, *scared*) and high EDA confusion (*discomfort*, *hungry*) show the weakest per-class F1 scores, confirming the EDA predictions.

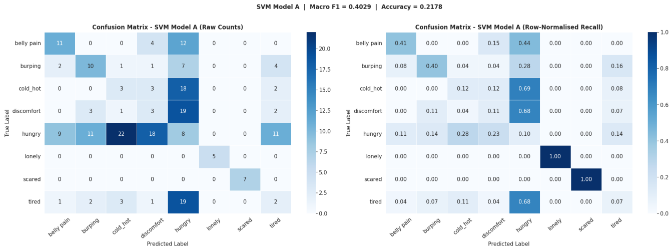

Fig. 5: Row-normalised confusion matrix for SVM Model A on the test set. Each cell shows the fraction of true-class samples predicted as the column class. Darker diagonal cells indicate higher recall. Off-diagonal concentration in the *hungry* column confirms the model's majority-class bias despite balanced class weights.

TABLE VI: Per-Class SVM Results on Test Set

Class	Prec.	Rec.	F1
belly pain	0.4783	0.4074	0.4400
burping	0.3846	0.4000	0.3922
cold_hot	0.1000	0.1154	0.1071
discomfort	0.1000	0.1071	0.1034
hungry	0.0964	0.1013	0.0988
lonely	1.0000	1.0000	1.0000
scared	1.0000	1.0000	1.0000
tired	0.0952	0.0714	0.0816
Macro avg	0.4068	0.4003	**0.4029**

Fig. 6 shows per-class recall and the top confused pairs ranked by test error count. The three most-confused pairs match exactly the EDA nearest-neighbour predictions (Section III-D), confirming that failures stem from genuine acoustic similarity rather than model weakness alone. We explain each mathematically below.

E. Failure Analysis

1) Failure Mode 1: discomfort ↔ hungry: Both classes exhibit rhythmic cry bursts at fundamental frequency $F_0 \in [300, 500]$ Hz with harmonics at $2F_0, 3F_0, \ldots$. The perceptual difference lies in inter-burst periodicity, which manifests as local spectro-temporal patterns detectable by convolutional filters but lost upon flattening. Formally, two spectrograms differing only in cry-onset offset δ frames produce:

$$\|\mathbf{x} - \mathbf{x}'\|^2 = \sum_{m=1}^{128} \sum_{t=1}^{302} \big(S[m, t] - S[m, t - \delta]\big)^2 \tag{10}$$

This distance can be large even for acoustically identical cries, driving the RBF kernel toward zero (dissimilar). The SVM thus classifies recordings by timing artefacts rather than acoustic content.

2) Failure Mode 2: hungry ↔ tired: Tired cries exhibit a gradually declining F_0 over time. However, peak normalisation (Eq. 1) removes the absolute energy difference that separates these classes perceptually. After normalisation, distinguishing them requires tracking the F_0 decay trajectory across 302 time frames — a sequential pattern that the static RBF kernel cannot learn from a flattened vector.

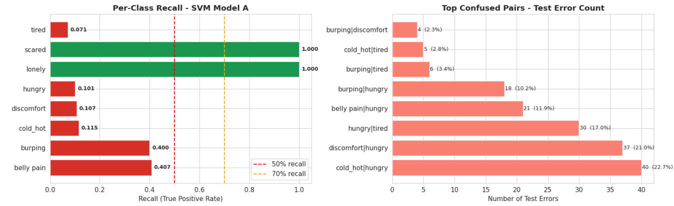

Fig. 6: Left: per-class recall for SVM Model A. Red bars (<50%) indicate critical failure classes; orange bars (50–70%) indicate weak performance. Right: top confused pairs ranked by raw test error count, with percentage of total errors annotated. The top-3 pairs match the EDA nearest-neighbour overlap predictions.

3) Failure Mode 3: cold_hot ↔ hungry: Cold_hot cries are typically shorter (4–5 s). After 7-second padding, these clips contain zero-valued silence in approximately frames 190–302. After StandardScaler normalisation, the silent region has near-zero variance $\sigma_k \approx 0$, producing a very large scaling factor $1/\sigma_k \gg 1$ that amplifies noise in these dimensions. The RBF kernel interprets the inflated distance as evidence of separability — but the learned boundary reflects recording length artefacts, not acoustic content, and does not generalise.

4) Architectural Ceiling: The three failure modes share a common root cause: the SVM on flattened inputs cannot (1) learn local spectro-temporal patterns (requires 2-D convolution), (2) achieve translation invariance (requires max-pooling), or (3) exploit temporal ordering (requires temporal convolution or recurrence). These are architectural limitations, not data limitations. The Depthwise Separable CNN (Phase 2) addresses all three by treating the $(128, 302)$ spectrogram as a 2-D image.

VII. Conclusion and Phase 2 Preview

This paper establishes the complete Phase 1 foundation for the Infant State Recognition System. Key contributions include the identification and correction of a sampling-rate confound in the Baby Cry Sense Dataset, a fully reproducible preprocessing and feature engineering pipeline producing 1,334 Log-Mel Spectrograms of shape $(128, 302)$, and a rigorously analysed SVM baseline achieving macro F1 = 0.4029. The failure analysis provides three mathematically grounded explanations for the SVM ceiling, each directly motivating a specific architectural choice in Phase 2.

Phase 2 will introduce a Depthwise Separable CNN (Model B) operating directly on the $(128, 302)$ spectrogram image. The DS-CNN applies $\frac{1}{N} + \frac{1}{D_K^2}$ fewer multiplications than a standard convolution [7], enabling TensorFlow Lite Micro deployment on the ESP32. Grad-CAM interpretability will verify that the model attends to genuine cry frequency bands. Phase 3 will fuse Models A and B into a hybrid system (Model C) that must definitively outperform both baselines on macro F1-score across all 8 classes.

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